

財政學 HW3

1. (Hyman 3.3) The EPA wants to reduce emissions of sulfur dioxides from electric power-generating plants by 20 percent during the next year. To achieve this goal, compensate receptors of external damage in cases where there are few emitters and many receptors?

the EPA will require each power-generating plant in the nation to reduce emissions by 100 tons per year. Suppose five power plants emit sulfur dioxides and serve a given metropolitan area. The following table shows the cost per ton of reducing emissions for each of the five plants:

COST PER TON OF PLANT	EMISSIONS REDUCTION (\$)
1	600
2	500
3	500
4	400
5	200

Assuming that the cost per ton of emissions reduction is constant and that the improvement in the air for the metropolitan area is the same no matter which plant reduces emissions, calculate the following:

- Cost of meeting EPA regulations.
- Least-cost method of achieving the EPA goal of reducing emissions of sulfur dioxides from power plants in the metropolitan area

2. (Hyman 3.4) Instead of using regulations to achieve the 20 percent reduction in emissions discussed in the preceding problem, suppose the EPA requires each of the five emitters to pay a fee of \$450 for each ton of sulfur dioxide it dumps in the air during the year.

Use the data from the table for problem 1 to predict which companies will purchase pollution rights, total cost of achieving the reduction in sulfur dioxide emissions, and revenue generated from the sale of pollution rights in the area.

3. (Rosen 5.7) For each of the following situations, is the Coase Theorem applicable? Why or why not?

- a. A farmer who grows organic corn is at risk of having his crop contaminated by genetically modified corn grown by his neighbors.
- b. In Silveiras, Brazil, queen ants are regarded as a delicacy, but recently the ant haul has been dwindling because of pesticides used on eucalyptus trees that are planted to produce cellulose for paper and other products [Barrionuevo and Demit, 2011]
- c. Following a military coup in Madagascar, a weak president was installed who was unable to stop the selling of Rosewood trees to China that were illegally cut down from the national parks.
- d. User of the Internet generally incur a zero incremental cost for transmitting information. As a consequence, congestion occurs, and users are frustrated by delays.

4. (Rosen 5.9) In India, a drug used to treat sick cows is leading to the death of many vultures that feed off of dead cattle. Before the decrease in the number of vultures, they sometimes used to smash into the engines of jets taking off from New Delhi's airports, posing a serious threat to air travelers. However, the decline of the vulture population has led to a sharp increase in the populations of rats and feral dogs, which are now the main scavengers of rotting meat [Gentleman, 2006, p. A4]. There have been calls for a ban on the drug used to treat the cows.

Identify the externalities that are present in this situation. Comment on the efficiency of banning the drug. How would you design an incentive-based regulation to attain an efficient outcome

5. (Rosen 5.11) American suburbs are expanding to more rural areas at the same time as a pig farms are expanding in size [*Economist*, 2007d, p. 36]. The smells emanating from the massive amounts of pig manure adversely affect property values. Imagine that the Little Pigs (LP) hog farm is situated near 100 houses. The following table shows, for each level of LP's output, the marginal cost (MC) of a hog, the marginal benefit (MB) to LP, and the marginal damage (MD) done to property values:

Output	MC	MB	MD
1	400	1600	400
2	800	1600	800
3	1200	1600	900
4	1600	1600	1000
5	3200	1600	1200
6	6400	1600	1400

- How many hogs does LP produce?
- What is the efficient number of hogs?
- Suppose the owner of LP can reduce the marginal damages of hog smells by two-thirds by modifying the hogs' diet. The modified diet increases the marginal cost of each hog by \$100. What is the efficient number of hogs?

6. 閱讀老師稅評「政府關門？總預算案僵局的真相看過來」，文章中有非常多法條值得留意，能幫助同學們更了解總預算案的架構，請根據題目敘述找出相對應的法條。（共有7小題，第a小題為範例）

- 根據預算法，總預算案應於會計年度開始一個月前由立法院完成審議（議決），並於會計年度開始15日前由總統公布。

答:預算法第51條

- 根據立法院職權行使法，預算案應經三讀會議決。

答:

- 根據預算法，總預算案未能於期限內完成議決，各機關預算之執行，按「馬照跑、舞照跳」的原則處理；簡單來說，「該收的稅費照收」、「該發的金錢照發」、「延續中計畫照做」、「必須借的錢照借」，是以絕對不會出現政府關門的情形。在「馬照跑、舞照跳」原則下，唯有「新增計畫」與「新興資本支出」，須俟預算案完成審議程序後，始得動支。

答:

- 根據預算法，新增計畫與新興資本支出，在預算案編列的額度內，只要立法院同意，仍可執行！

答:

- 行政院訂有中央政府總預算案未能依限完成時之執行補充規定的行政規章，正是為了避免總預算案之審議，因故全部或部分未能通過，而使政府施政中輟。

答:

f. 根據憲法，立法院會期，每年兩次。

答：

g. 根據憲法增修條文，每屆立委任期四年，故共有八個會期。

答：